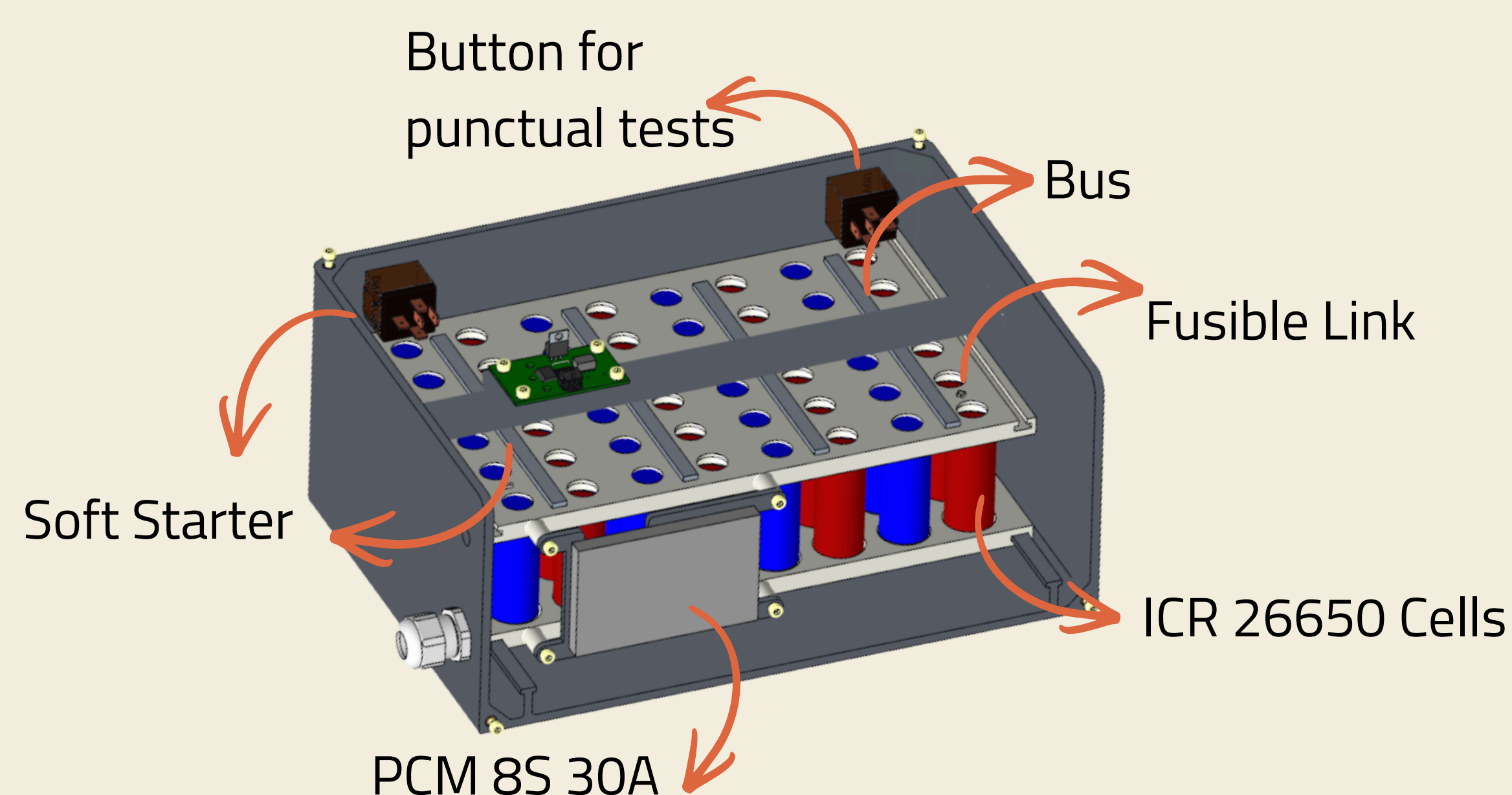


LV

Battery

Battery Configuration: 8s 6p

- ICR 26650
- Nominal voltage: 3.7V
- Maximum voltage: 4.2V
- Maximum current: 15A
- Maximum battery voltage: 33.6V
- Parallel Cells protected by fusible link



LV

Decision matrix for cell selection

Ideal cell	Parameter	Last cell Li1x7pHG2	ICR26650	Li ion - Sanyo	Li ion - Samsung
Time: 1,5	Mass (kg)	2688	4416	3024	3360
Energy (Wh): 656,66	Energy (Wh)	604,8	888	725,76	691,2
Capacity: 19,89	Cost	Sponsored	1930,94	2988,79	5772
	Reasons	Discharge fast	↑ Energy ↓ Cost	↓ Mass	Worst option

LV

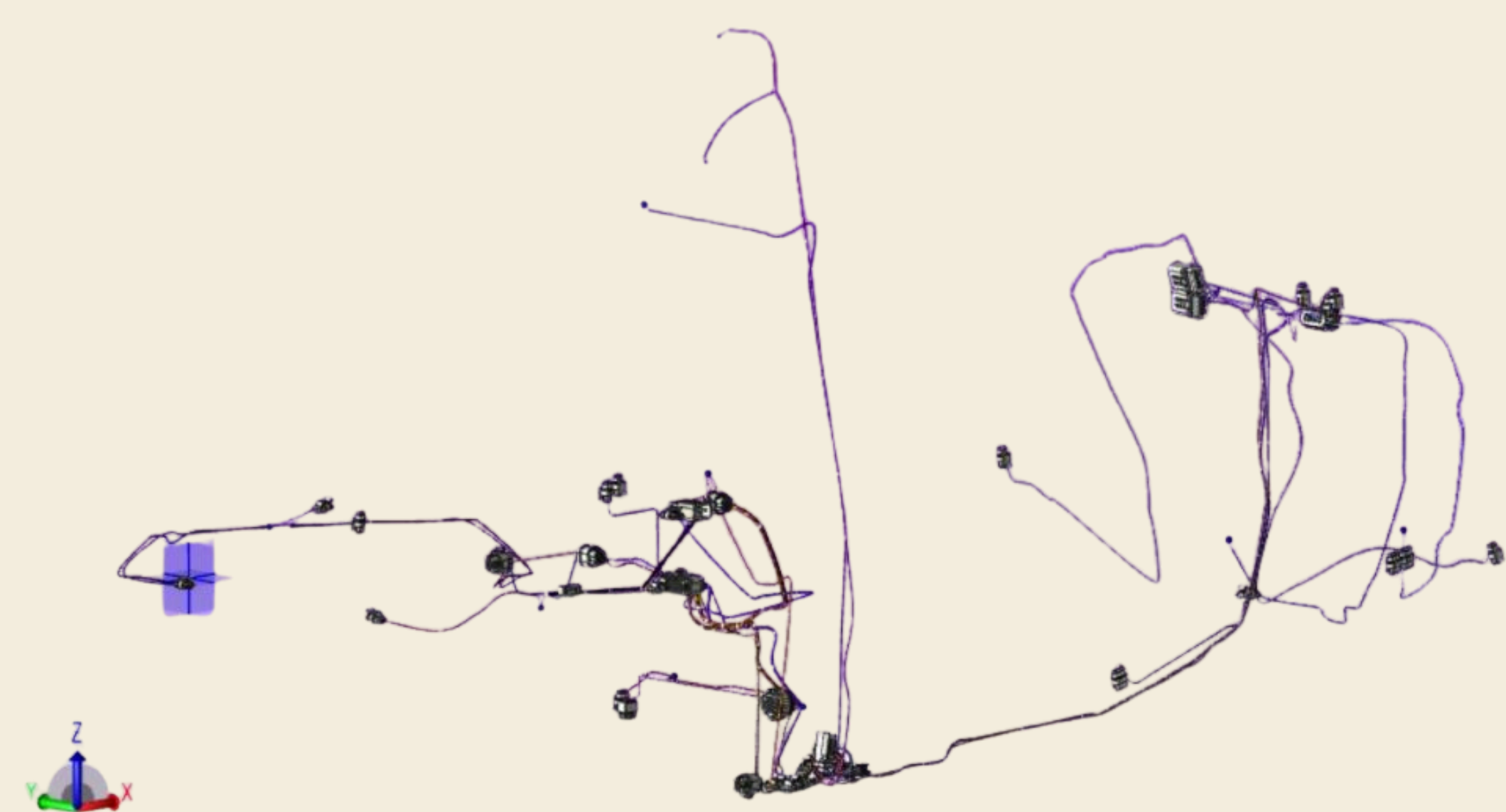
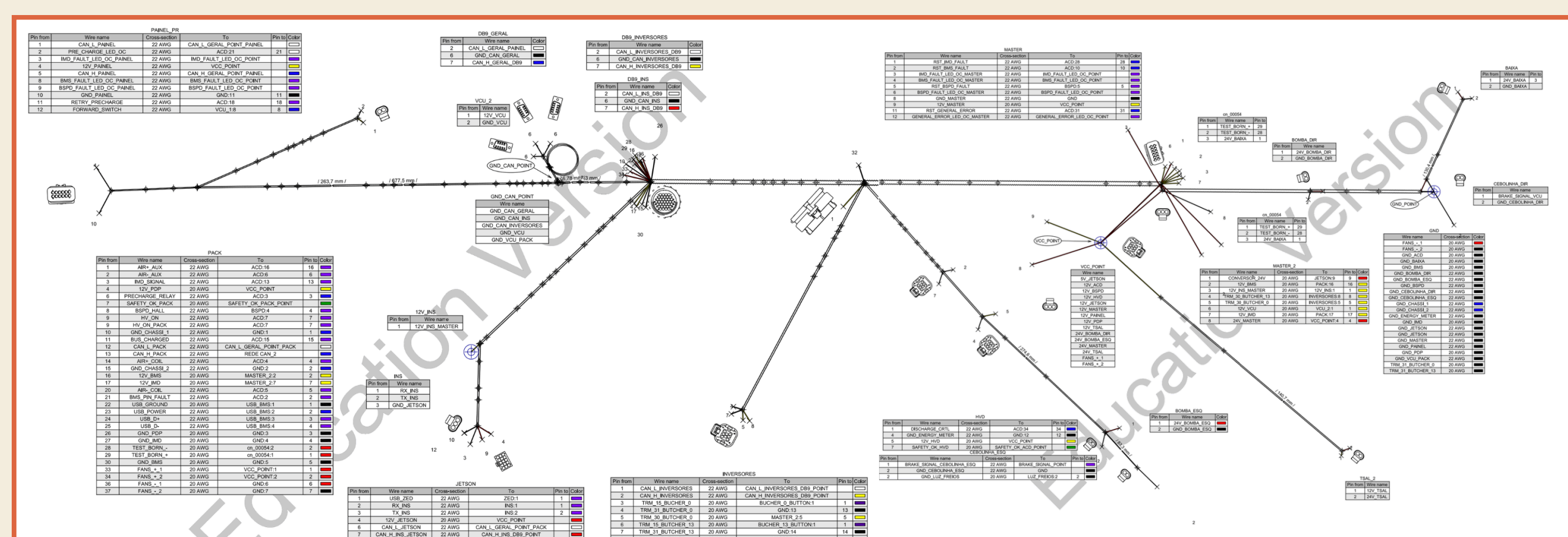
Wiring Harness

Wires

- Color coded
- Wire gauges from 18 AWG to 22 AWG
- Faraday tape

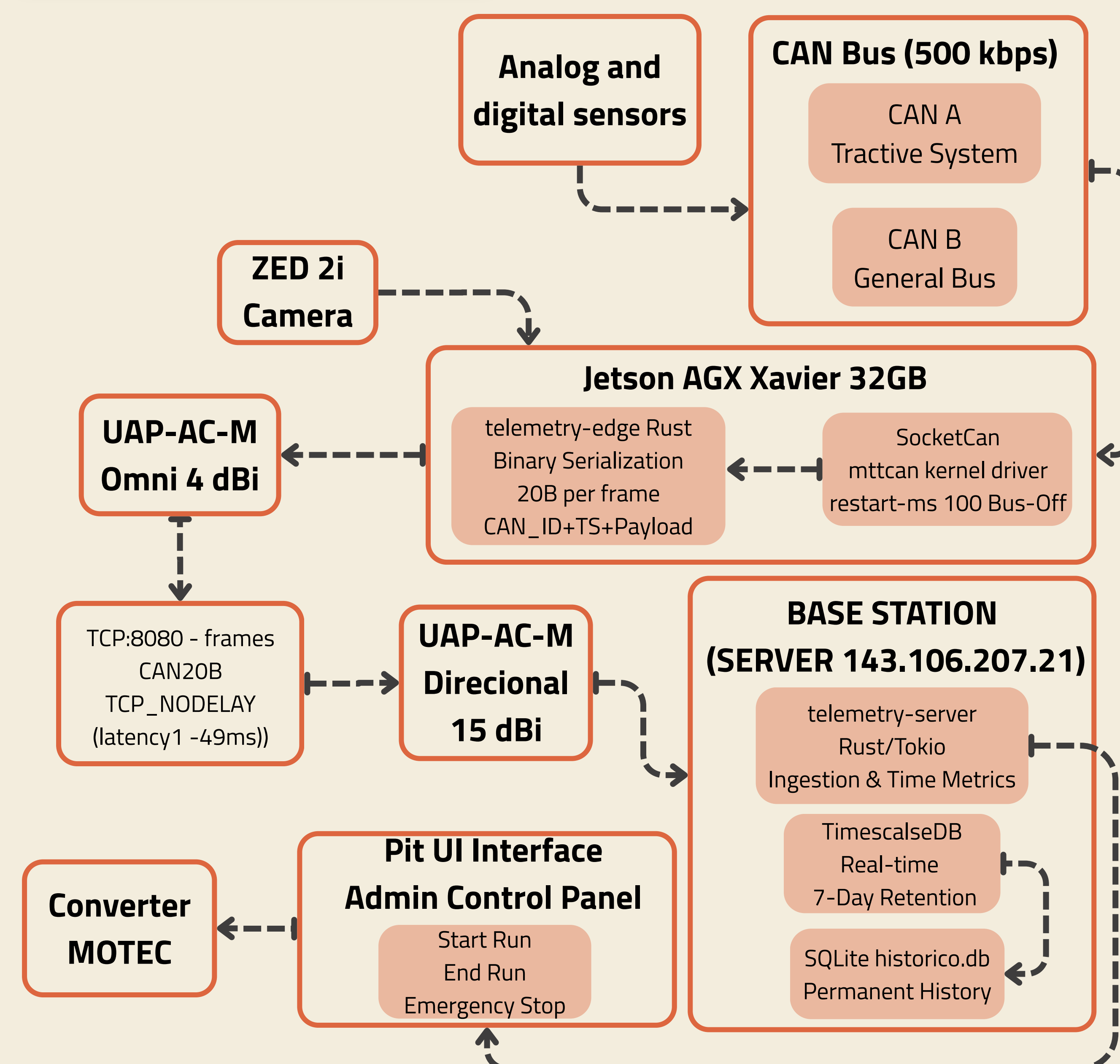
Connectors

- Criteria: Water resistance, positive-locking



Data Acq.

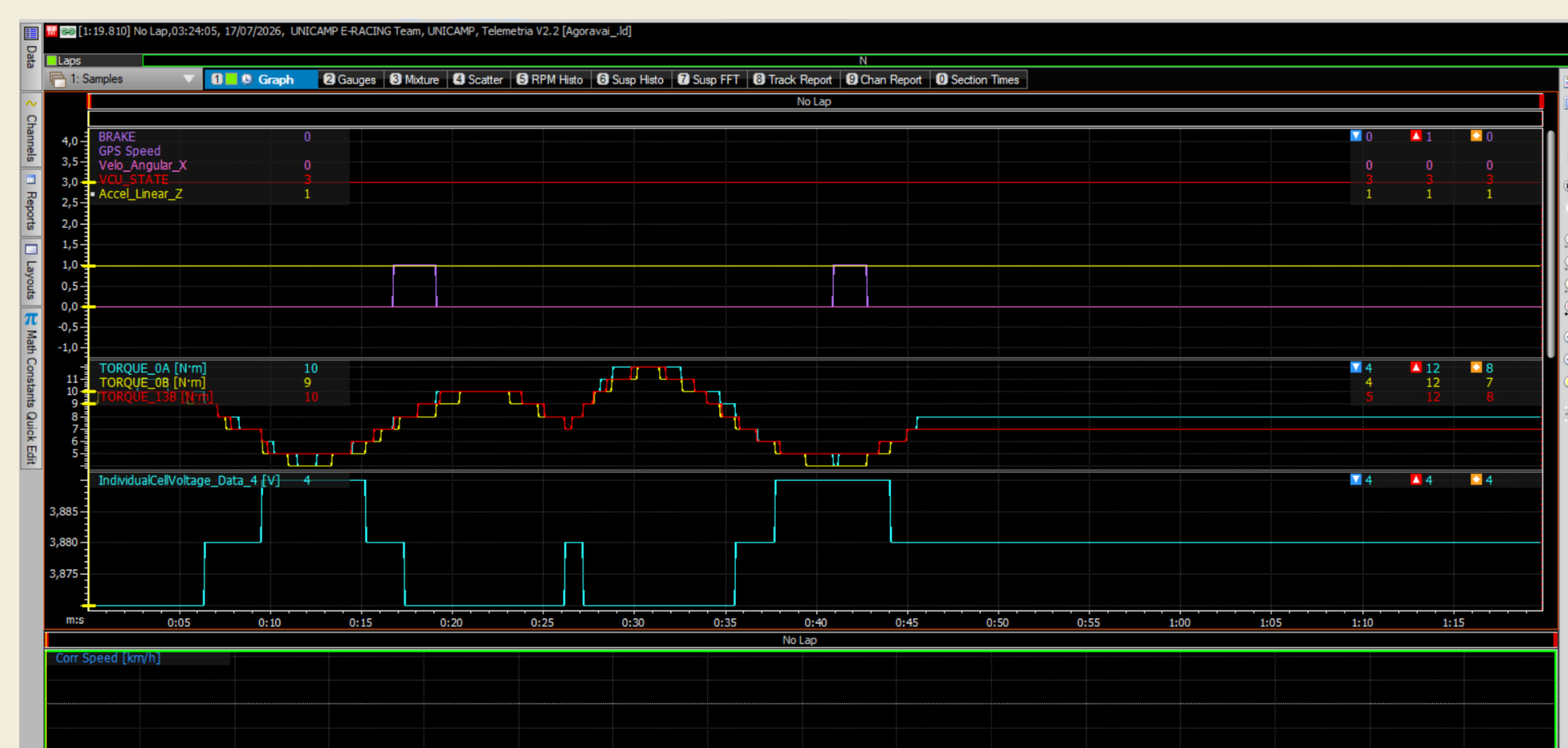
Data and Telemetry



Data Acq.

Data Analysis: Motec

- Fully implemented in the telemetry pipeline
- Crucial for performance decisions



Data Acq.

AI Performance Bench Benchmarking

- SLAM (ROS2 + Cartographer)
- MLT (Minimum Lap Time) solver
- Driver-in-the-Loop (DIL)
 - Physics-Based Simulation (Track + Vehicle)
 - Vehicle Model
 - VCU Control-in-the-Loop
- Reinforcement Learning Pilot

